

Our analysis is structured as follows. We introduce the correct functional analytic setup in Section 3, where we also prove the Fréchet differentiability of $\mathcal{J}$ and $\mathcal{C}$ and give formulas for their derivatives. In Section 4 we provide the proof of Theorem 2.1.

3. ANALYSIS OF THE FUNCTIONAL $\mathcal{F}$

In this section, we study the regularity properties of the functional $\mathcal{F}$ and derive and analyze its linearization. First, we introduce the underlying function spaces for functions $\eta: \mathbb{S}^2 \rightarrow \mathbb{R}$ and recall some important facts on Sobolev spaces on the sphere.

3.1. Sobolev spaces on the sphere. We denote by $L^2(\mathbb{S}^2)$ the space of square-integrable functions on the sphere equipped with the uniform measure $d\sigma(x) = \sin(\theta) d\varphi d\theta$ (which is the same as the Hausdorff measure on $\mathbb{S}^2$ up to a constant factor) and we write $\langle \cdot, \cdot \rangle$ for the induced L^2 scalar product.

Our analysis is based on the fact that spherical harmonics $\{Y_l^m : l \in \mathbb{N}_0, -l \leq m \leq l\}$ form an orthonormal eigenbasis of the Laplace–Beltrami operator on the sphere $\Delta_{\mathbb{S}^2}$ with respect to the $L^2(\mathbb{S}^2)$ scalar product. The corresponding eigenvalues $-l(l+1)$ have the multiplicity $2l+1$. We have the expansion

$$f = \sum_{l=0}^{\infty} \sum_{m=-l}^l \langle f, Y_l^m \rangle Y_l^m, \quad (3.1)$$

for any $f \in L^2(\mathbb{S}^2)$. We recall that spherical harmonics can be expressed as

$$Y_l^m(\theta, \varphi) = c_{l,m} P_l^m(\cos \theta) e^{im\varphi}, \quad (3.2)$$

where $c_{l,m} = \sqrt{\frac{(2l+1)}{4\pi} \frac{(l-m)!}{(l+m)!}}$ are positive constants and P_l^m are the associated Legendre polynomials. We refer the reader e.g. to [44] for background reading.

For $\beta > 0$, we define the Sobolev space $H^\beta(\mathbb{S}^2)$ as the space of all functions $f \in L^2(\mathbb{S}^2)$ with

$$\|f\|_{H^\beta(\mathbb{S}^2)}^2 = \sum_{l=0}^{\infty} \sum_{m=-l}^l (1+l)^{2\beta} |\langle f, Y_l^m \rangle|^2 < \infty.$$

These spaces can be equivalently defined via smooth charts [27], and thus, they arise as the trace spaces of $H^{\beta+\frac{1}{2}}(B_1(0))$, cf. (3.8). For integer exponents $\beta \in \mathbb{N}$, they coincide with the classical Sobolev spaces defined via differentiation on the manifold.

For notational convenience, we introduce a subspace of $H^\beta(\mathbb{S}^2)$ that reflects the symmetric setting we restrict to in (1.23).

Definition 3.1. Let $\beta \geq 0$. We define

$$H_{\text{sym}}^\beta(\mathbb{S}^2) := \left\{ f \in H^\beta(\mathbb{S}^2) : f = f(\theta) \text{ with } f\left(\frac{\pi}{2} - \theta\right) = f\left(\frac{\pi}{2} + \theta\right) \right\},$$

the subspace of all axisymmetric functions in $H^\beta(\mathbb{S}^2)$, which are symmetric in x_3 .

The following characterization will be beneficial for our analysis.

Lemma 3.2. For all $\beta \geq 0$ we have

$$H_{\text{sym}}^\beta(\mathbb{S}^2) := \left\{ f \in H^\beta(\mathbb{S}^2) : \langle f, Y_l^m \rangle = 0 \text{ if } l \text{ is odd or } m \neq 0 \right\}. \quad (3.3)$$